

DB PROJECT PRESENTATION

By: Mohammed and Osama

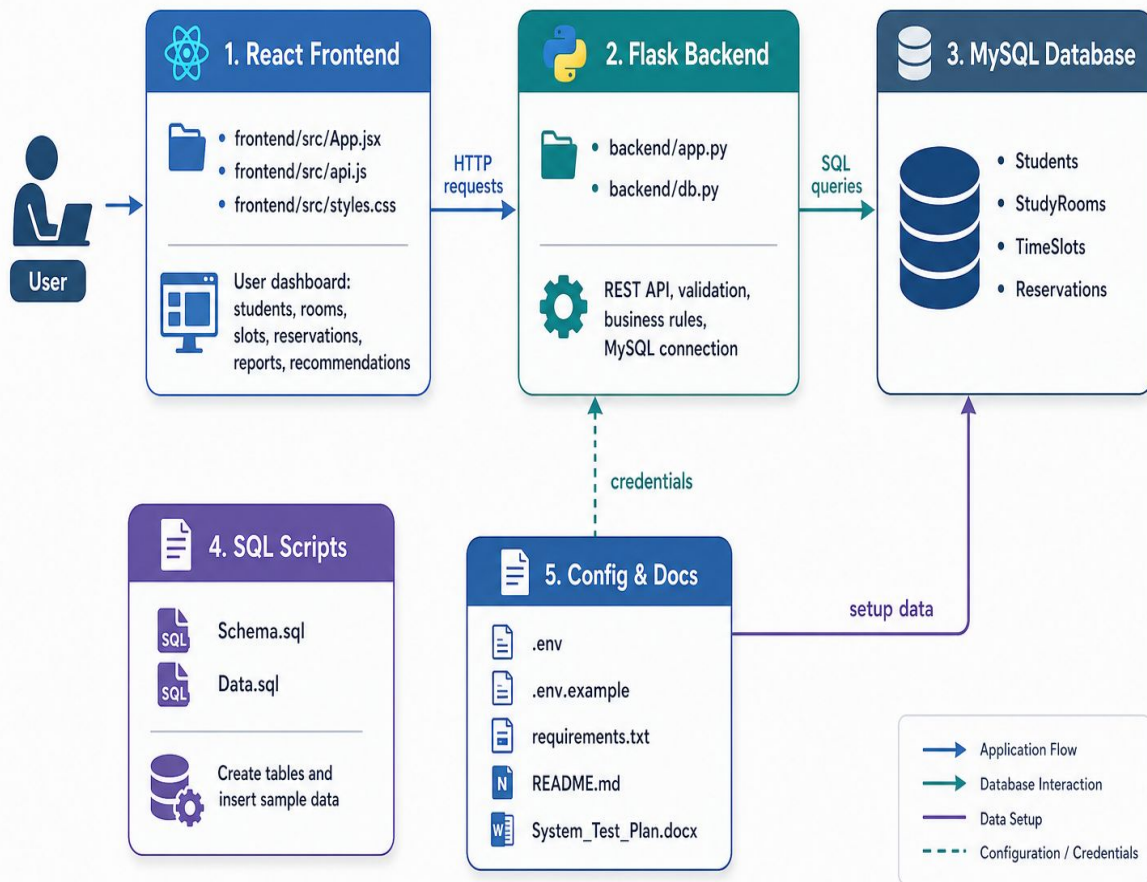
Campus Study Room Reservation System

- *Manages students, rooms, time slots, and reservations.*
- *Only active students can book rooms.*
- *Prevents double booking and invalid reservations.*
- *Recommends available rooms based on capacity, type, and time slot (Advanced Feature)*

SYSTEM ARCHITECTURE

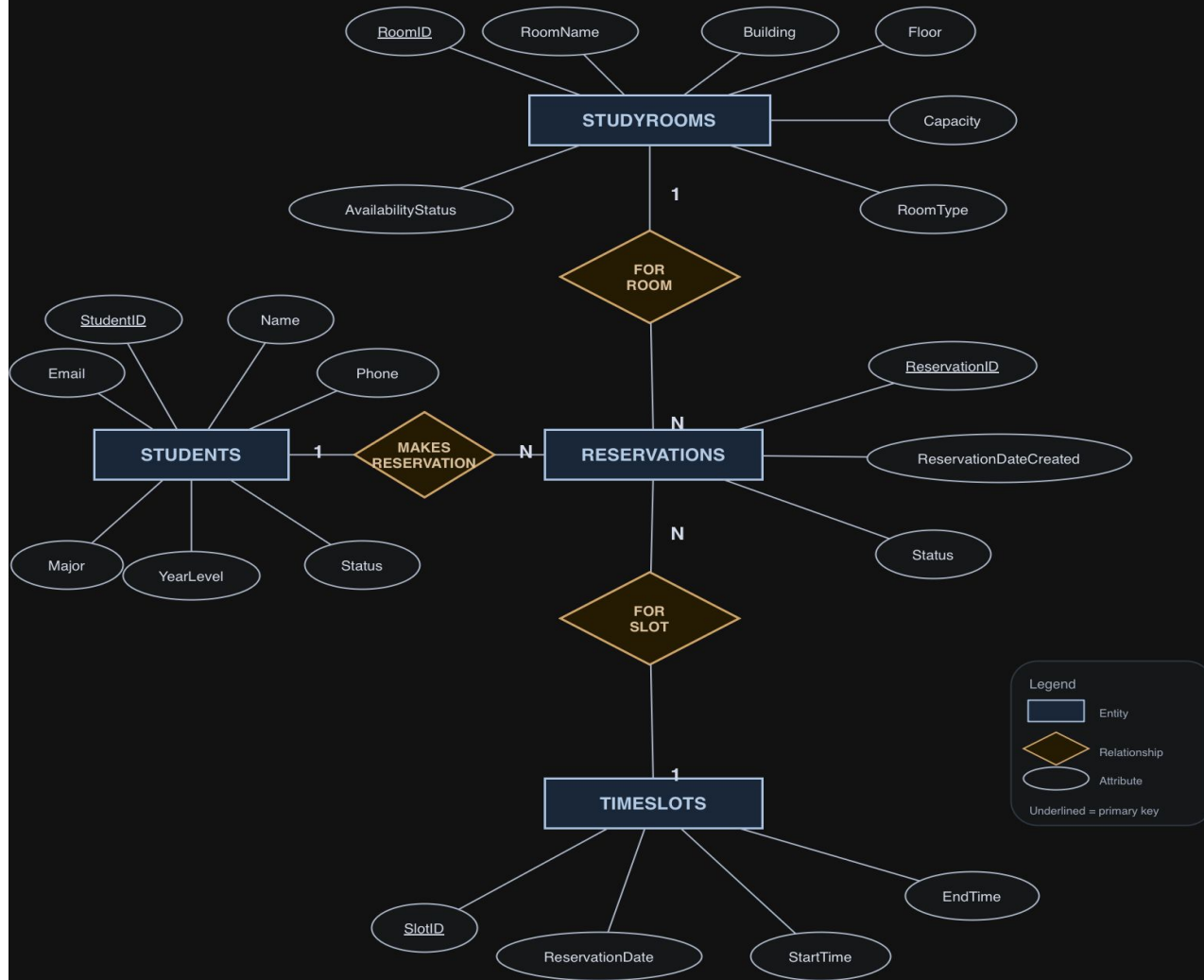
GENERAL STRUCTURE OF CODE

Campus Study Room Reservation System - Codebase Structure



ENTITY RELATION DIAGRAM

ER Diagram - Study Room Reservation System



RELATIONAL SCHEMA DIAGRAM

STUDENT

<u>StudentID</u>	Name	Email	Phone	Major	YearLevel	Status
------------------	------	-------	-------	-------	-----------	--------

RESERVATION

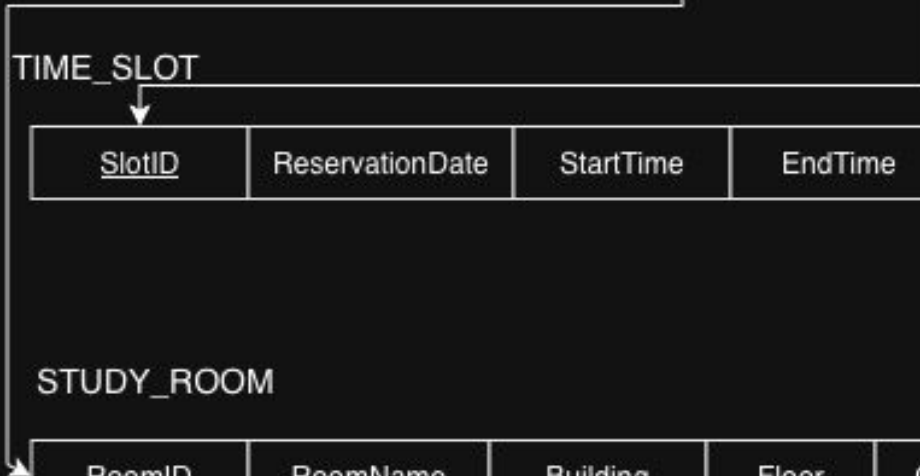
<u>ReservationID</u>	StudentID	RoomID	SlotID	ReservationDateCreated	Status
----------------------	-----------	--------	--------	------------------------	--------

TIME_SLOT

<u>SlotID</u>	ReservationDate	StartTime	EndTime
---------------	-----------------	-----------	---------

STUDY_ROOM

<u>RoomID</u>	RoomName	Building	Floor	Capacity	RoomType
---------------	----------	----------	-------	----------	----------



KEY FEATURES

STUDENT MANAGEMENT

STUDENTS

9

ROOMS

7

SLOTS

11

RESERVATIONS

13



Students



Rooms



Time Slots



Reservations



Reports



Advanced



Add Student

NAME

EMAIL

PHONE

MAJOR

YEAR

STATUS

Add Student



Update Or Deactivate

STUDENT ID

EMAIL

PHONE

MAJOR

YEAR

STATUS

Update

Deactivate



Search Students

ID

NAME

EMAIL

MAJOR

YEAR

STATUS

Search

STUDENTID	NAME	EMAIL	PHONE	MAJOR	YEARLEVEL	STATUS
1	Aisha Al Mansoori	aisha.almansoori@ku.ac.ae	+971501112233	Computer Science	3	Active
2	Omar Haddad	omar.haddad@ku.ac.ae	+971502223344	Electrical Engineering	2	Active

STUDY ROOM MANAGEMENT

STUDENTS
9

ROOMS
7

SLOTS
11

RESERVATIONS
13

- Students
- Rooms
- Time Slots
- Reservations
- Reports
- Advanced

Add Study Room

ROOM NAME

BUILDING

FLOOR

CAPACITY

TYPE

AVAILABILITY

Add Room

Update Or Remove

ROOM ID

BUILDING

FLOOR

CAPACITY

TYPE

AVAILABILITY

Update

Mark Unavailable

Search Rooms

ID

NAME

BUILDING

FLOOR

CAPACITY

TYPE

Search

ROOMID	ROOMNAME	BUILDING	FLOOR	CAPACITY	ROOMTYPE	AVAILABILITYSTATUS
7	OS-AMA	Building L	2	3	Group	Available
3	SR-210	Engineering Building	2	6	Group	Available

TIME SLOT MANAGEMENT

STUDENTS

9

ROOMS

7

SLOTS

11

RESERVATIONS

13

Students

Rooms

Time Slots

Reservations

Reports

Advanced

📅 Add Time Slot

DATE

04/28/2026

START

12:30 PM

END

12:30 PM

Add Slot

🔍 Search Time Slots

DATE

04/28/2026

START

12:30 PM

END

12:30 PM

FUTURE

All

Search

📅 Time Slots

SLOTID	RESERVATIONDATE	STARTTIME	ENDTIME
9	2026-04-25	13:00:00	14:00:00
10	2026-04-26	16:00:00	17:30:00
1	2026-04-28	09:00:00	10:00:00
12	2026-04-28	09:30:00	10:00:00
2	2026-04-28	10:00:00	11:00:00
3	2026-04-28	11:00:00	12:00:00
4	2026-04-29	09:00:00	10:00:00
5	2026-04-29	10:00:00	11:00:00

RESERVATION MANAGEMENT

STUDENTS
9

ROOMS
7

SLOTS
11

RESERVATIONS
13

Students

Rooms

Time Slots

Reservations

Reports

Advanced

✓ Reserve A Room

STUDENT ID

ROOM ID

SLOT ID

Reserve

Reservation Actions

RESERVATION ID

Check In

Cancel

Complete

Search Reservations

STUDENT	ROOM	DATE	SLOT	STATUS				
		04/28/2026		Any	Search			
RESERVATIONID	STUDENTNAME	ROOMNAME	BUILDING	SLOTID	RESERVATIONDATE	STARTTIME	ENDTIME	STATUS
12	Daniel Martins	SR-305	Innovation Center	8	2026-04-30	15:00:00	16:00:00	Completed
6	Mariam Saeed	SR-410	Science Building	5	2026-04-29	10:00:00	11:00:00	Reserved
5	Yousef Khan	Quiet Pod A	Main Library	4	2026-04-29	09:00:00	10:00:00	CheckedIn
9	Noura Salem	SR-102	Main Library	3	2026-04-28	11:00:00	12:00:00	Reserved
0	Daniel Martins	SR-305	Main Library	0	2026-04-28	11:00:00	12:00:00	Completed

SEARCH AND REPORT FUNCTIONALITY

Students

Rooms

Time Slots

Reservations

Reports

Advanced

Student Schedule

STUDENT ID

1

View Schedule

ENTRYTYPE	STATUS	ROOMNAME	BUILDING	RESERVATIONDATE	STARTTIME	ENDTIME
Reservation	Reserved	SR-101	Main Library	2026-04-28	09:00:00	10:00:00
Reservation	Completed	OS-AMA	Building L	2026-04-28	09:00:00	10:00:00
Reservation	Reserved	SR-210	Engineering Building	2026-04-28	10:00:00	11:00:00

Available Rooms

DATE

04/28/2026

SLOT ID

1

Find Rooms

ROOMID	ROOMNAME	BUILDING	FLOOR	CAPACITY	ROOMTYPE	AVAILABILITYSTATUS
5	Quiet Pod A	Main Library	2	1	Individual	Available
7	OS-AMA	Building L	2	3	Group	Available
3	SR-210	Engineering Building	2	6	Group	Available
4	SR-305	Innovation Center	3	8	Group	Available
6	SR-410	Science Building	4	10	Group	Available

Students For Room And Slot

ROOM ID

1

SLOT ID

1

View Students

STUDENTID	NAME	EMAIL	RESERVATIONID	STATUS	RESERVATIONDATE	STARTTIME	ENDTIME
1	Aisha Al Mansoori	aisha.almansoori@ku.ac.ae	1	Reserved	2026-04-28	09:00:00	10:00:00

ADANCED FEATURE

ROOM AVAILABILITY RECOMMENDATION

COSC444 DATABASE SYSTEMS

Campus Study Room Reservation System



STUDENTS
9

ROOMS
7

SLOTS
11

RESERVATIONS
13

- Students

Rooms

Time Slots

Reservations

Reports

Advanced

Room Recommendation

DATE

04/28/2026

SLOT ID

1

CAPACITY

3

TYPE

Group

SORT

Capacity

Recommend

ROOMID	ROOMNAME	BUILDING	FLOOR	CAPACITY	ROOMTYPE	RESERVATIONDATE	STATUS
7	OS-AMA	Building L	2	3	Group	2026-04-28	09
3	SR-210	Engineering Building	2	6	Group	2026-04-28	09
4	SR-305	Innovation Center	3	8	Group	2026-04-28	09
6	SR-410	Science Building	4	10	Group	2026-04-28	09

DESIGN CHALLENGES

- Preventing double booking was challenging because two students could try to reserve the same room and time slot.

What we did: The backend checks existing active reservations before creating a new one and blocks the request if the room is already reserved.

- Following all reservation rules was challenging because a booking depends on multiple conditions: active student, available room, and valid future time slot.

What we did: We added validation before inserting a reservation so invalid bookings are rejected early. Validate (through SQL calls) they are active, the room exists, and the slot is not taken, before inserting reservation.